ASSIGNMENT : 2

TITLE : Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Problem Statement : Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives : To understand and implement default, parameterized, and copy constructors for object initialization.

Theory : Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Exercise :

1. Create a **Student class** using default and parameterized constructors to initialize student name and roll number.

```java
class Student {
    String name;
    int roll;
    int age;

    Student() {
        name = "Anushree";
        roll = 9;
        age = 18; }

    Student(String n, int r, int a) {
        name = n;
        roll = r;
        age = a; }

    void display() {
        System.out.println("Name: " + name + " | PRN: " + roll + " | Age: " + age);
    }
}

public class Main {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.display();
        Student s2 = new Student("Anish",13,17);
        s2.display();
    }
}
```

```
Output
Name: Anushree | PRN: 9 | Age: 18
Name: Anish | PRN: 13 | Age: 17
```

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

```java
class Mobile {
    String model;
    int price;

    Mobile(){
        model = "Samsung Galaxy A15";
        price = 20000;
    }

    Mobile(String m, int p){
        model = m;
        price = p;
    }

    Mobile( Mobile obj){
        model = obj.model;
        price = obj.price;
    }

    void display() {
        System.out.println("Model : " + model);
        System.out.println("Price :" + price);
        System.out.println();
    }
}

public class Main {
    public static void main(String[] args) {

        Mobile m1 = new Mobile();
        Mobile m2 = new Mobile("Apple iPhone 16", 85000);
        Mobile m3 = new Mobile(m2);

        System.out.println("Mobile 1 Details:");
        m1.display();
        System.out.println("Mobile 2 Details:");
        m2.display();
        System.out.println("Duplicate Mobile Record:");
        m3.display();
    }
}
```

```
Output
Mobile 1 Details:
Model : Samsung Galaxy A15
Price :20000

Mobile 2 Details:
Model : Apple iPhone 16
Price :85000

Duplicate Mobile Record:
Model : Apple iPhone 16
Price :85000
```

Conclusion

This assignment demonstrated the use of default, parameterized, and copy constructors in Java for object initialization and data management. The default constructor initialized objects with predefined values, the parameterized constructor allowed objects to be created with user-defined values, and the copy constructor created duplicate objects by copying the data of an existing object. These constructors simplify object creation, improve code readability, reduce repetitive initialization code, and support the principles of object-oriented programming.